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CpE 418: SOFTWARE DESIGN
Laboratory #3

Problem Analysis & Requirements Definition
(Phase 3)

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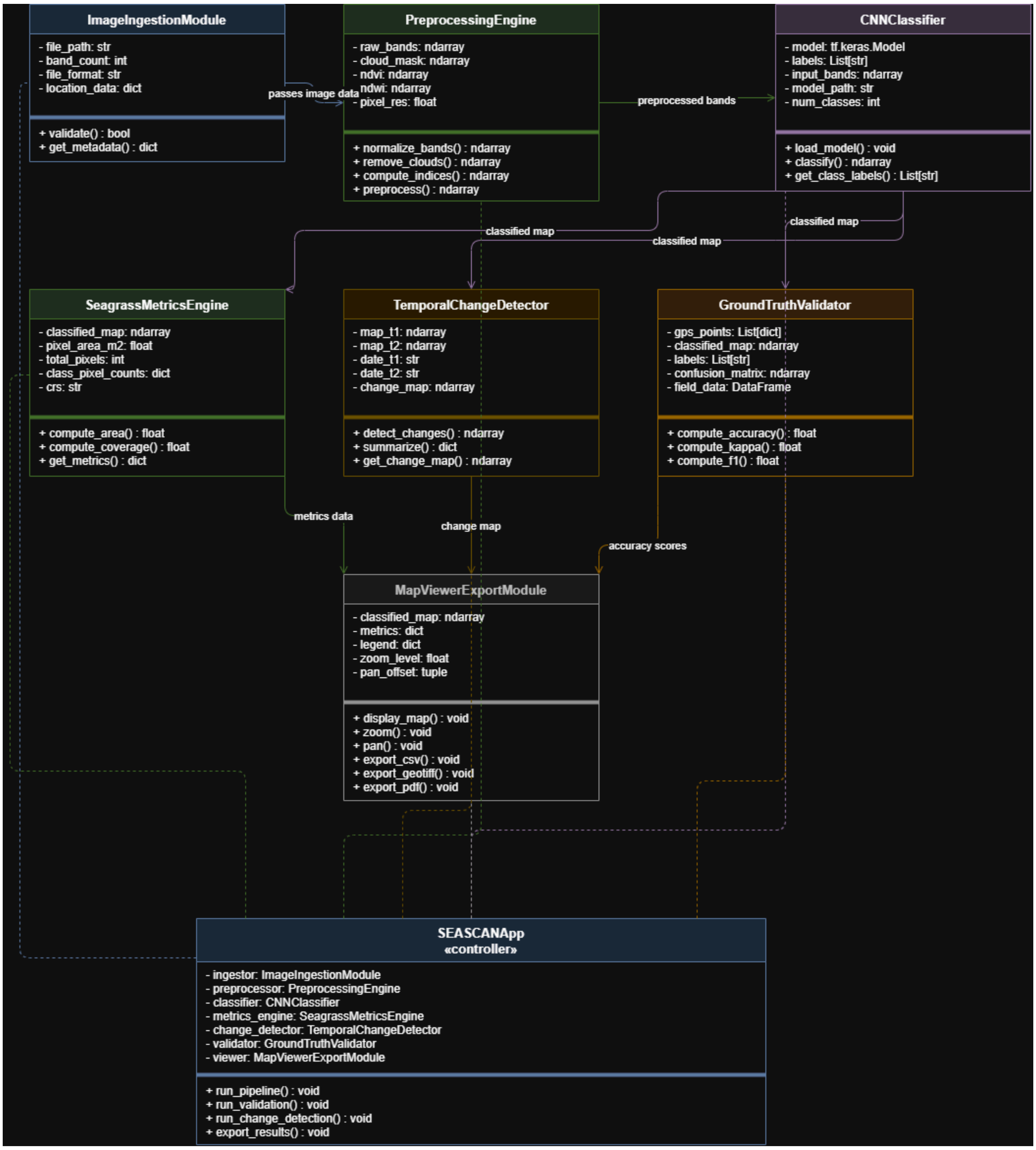
Section: CpE - 3202

Score: ____________________

Objective:

Laboratory Tasks:

Laboratory Output:


ImageIngestionModule
- file_path: str
- band_count: int
- file_format: str
- location_data: dict
+ validate() : bool
+ get_metadata() : dict
passes image data
PreprocessingEngine
- raw_bands: ndarray
- cloud_mask: ndarray
- ndvi: ndarray
- ndwi: ndarray
- pixel_res: float
+ normalize_bands() : ndarray
+ remove_clouds() : ndarray
+ compute_indices() : ndarray
+ preprocess() : ndarray
preprocessed bands
CNNClassifier
- model: tf.keras.Model
- labels: List[str]
- input_bands: ndarray
- model_path: str
- num_classes: int
+ load_model() : void
+ classify() : ndarray
+ get_class_labels() : List[str]
classified map
classified map
classified map
SeagrassMetricsEngine
- classified_map: ndarray
- pixel_area_m2: float
- total_pixels: int
- class_pixel_counts: dict
- crs: str
+ compute_area() : float
+ compute_coverage() : float
+ get_metrics() : dict
TemporalChangeDetector
- map_t1: ndarray
- map_t2: ndarray
- date_t1: str
- date_t2: str
- change_map: ndarray
+ detect_changes() : ndarray
+ summarize() : dict
+ get_change_map() : ndarray
GroundTruthValidator
- gps_points: List[dict]
- classified_map: ndarray
- labels: List[str]
- confusion_matrix: ndarray
- field_data: DataFrame
+ compute_accuracy() : float
+ compute_kappa() : float
+ compute_f1() : float
metrics data
change map
accuracy scores
MapViewerExportModule
- classified_map: ndarray
- metrics: dict
- legend: dict
- zoom_level: float
- pan_offset: tuple
+ display_map() : void
+ zoom() : void
+ pan() : void
+ export_csv() : void
+ export_geotiff() : void
+ export_pdf() : void
SEASCANApp
«controller»
- ingestor: ImageIngestionModule
- preprocessor: PreprocessingEngine
- classifier: CNNClassifier
- metrics_engine: SeagrassMetricsEngine
- change_detector: TemporalChangeDetector
- validator: GroundTruthValidator
- viewer: MapViewerExportModule
+ run_pipeline() : void
+ run_validation() : void
+ run_change_detection() : void
+ export_results() : void

SEASCAN System — Class Diagram Explanation			
Class	**Attributes**	**Methods**	**Description**
ImageIngestionModule	- file_path: str - band_count: int - file_format: str - location_data: dict	+ validate() : bool + get_metadata() : dict	Serves as the entry point of the system. It receives the uploaded Sentinel-2 image, verifies its format, band count, and location data, and returns an error before further processing if validation fails.
PreprocessingEngine	- raw_bands: ndarray - cloud_mask: ndarray - ndvi: ndarray - ndwi: ndarray - pixel_res: float	+ normalize_bands() : ndarray + remove_clouds() : ndarray + compute_indices() : ndarray + preprocess() : ndarray	Cleans and prepares the raw satellite image for classification. It adjusts band brightness values, removes cloud-covered pixels, and computes spectral indices (NDVI, NDWI) to help distinguish seagrass from other surface types.
CNNClassifier	- model: tf.keras.Model - labels: List[str] - input_bands: ndarray - model_path: str - num_classes: int	+ load_model() : void + classify() : ndarray + get_class_labels() : List[str]	Represents the core intelligence of the system. It loads a trained 1D Convolutional Neural Network and performs pixel-by-pixel classification, assigning each pixel one of five labels: seagrass, sand, land vegetation, open water, or cloud/shadow.

SeagrassMetricsEngine	- classified_map: ndarray - pixel_area_m2: float - total_pixels: int - class_pixel_counts: dict - crs: str	+ compute_area() : float + compute_coverage() : float + get_metrics() : dict	Converts the classified pixel map into real-world measurements. It calculates the total seagrass area in hectares and its percentage coverage relative to the scene, producing results directly usable in coastal resource reports.
TemporalChangeDetector	- map_t1: ndarray - map_t2: ndarray - date_t1: str - date_t2: str - change_map: ndarray	+ detect_changes() : ndarray + summarize() : dict + get_change_map() : ndarray	Compares classified maps from two different acquisition dates. It identifies pixels that changed class between dates and produces a change map with a summary indicating whether overall seagrass coverage increased or decreased, and by how much.
GroundTruthValidator	- gps_points: List[dict] - classified_map: ndarray - labels: List[str] - confusion_matrix: ndarray - field_data: DataFrame	+ compute_accuracy() : float + compute_kappa() : float + compute_f1() : float	An optional module used by researchers or marine biologists. It compares field-collected GPS observations against the classified map and computes Overall Accuracy, Kappa coefficient, and per-class F1 score to measure real-world model performance.

MapViewerExportModule	- classified_map: ndarray - metrics: dict - legend: dict - zoom_level: float - pan_offset: tuple	+ display_map() : void + zoom() : void + pan() : void + export_csv() : void + export_geotiff() : void + export_pdf() : void	The presentation layer of the system. It displays the classified map with a legend, allows the user to zoom and pan, and handles all output exports — producing a CSV of metrics, a GeoTIFF classified image, and a PDF summary report.
SEASCANApp «controller»	- ingestor: ImageIngestionModule - preprocessor: PreprocessingEngine - classifier: CNNClassifier - metrics_engine: SeagrassMetricsEngine - change_detector: TemporalChangeDetector - validator: GroundTruthValidator - viewer: MapViewerExportModule	+ run_pipeline() : void + run_validation() : void + run_change_detection() : void + export_results() : void	The top-level controller that orchestrates the entire system. It holds one instance of each of the seven module classes through aggregation and coordinates their sequential execution — from image ingestion through classification, analysis, and final export.

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